

SMRs without water the molten salt reactor, and all three

Precursor drift · freeze plugs and off-gas · MSRs at sea · a molten-salt system code · all three on one ruler

Week 6 · Instructor **Minseop Song** · Rev1.0 · September 2026

Parts 3 and 4 - five MSR chapters and two of synthesis

- Every reactor so far assumed **the fuel is solid and stays put**. The MSR drops that - and almost every textbook equation stood on it

12	The fuel flows	what changes without solid fuel
13	Circulating-fuel kinetics	delayed-neutron precursors drift out of the core
14	MSRs being built, and the sea	status in Sept 2026 - ages fastest
15	MSR safety	freeze plugs and off-gas
16	Molten-salt system code	the only one where properties are not enough
17	All three on one ruler	which cell of the property table stands out
18	Wrap-up and next steps	five equations and the open questions

12

The fuel flows

Fuel is dissolved in a fluoride or chloride salt and the salt itself circulates. The core is just a bundle of graphite channels that lets the liquid go critical. What follows from that one change is this whole chapter.

Dropping solid fuel - five gains, five costs

- **What needs active effort is staying on, not shutting down** - lose power and the plug melts and the fuel drops by gravity. The heart of the MSR passive-safety case

Gains

- 1 no cladding - no cladding failure, no Zr-water
- 2 no swelling or PCI - no burnup limit
- 3 atmospheric pressure - boils above 1400 °C
- 4 online refuelling and cleanup
- 5 a physical shutdown - the freeze plug

Costs

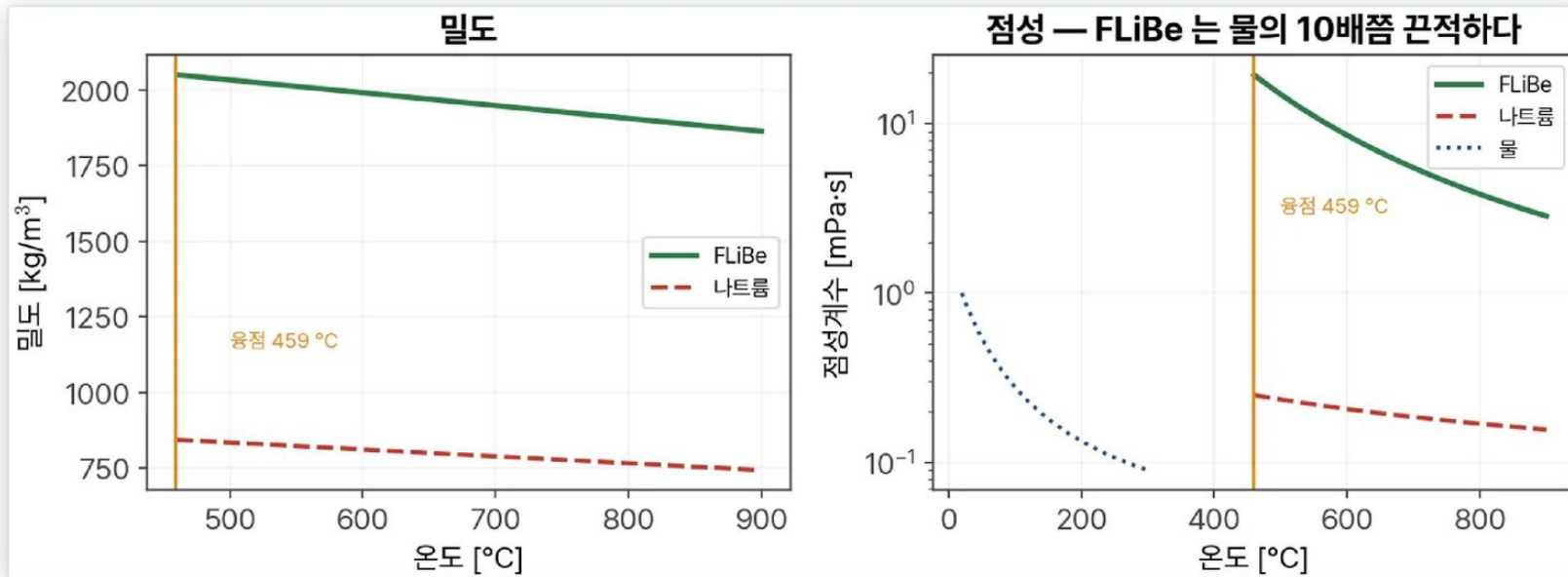
- 1 the whole primary is fuel boundary
- 2 precursors drift out of the core (ch. 13)
- 3 freezes below 459 °C - plugging
- 4 no qualified material - Hastelloy-N unlisted
- 5 chemistry is a safety system - redox

PCI pellet-cladding interaction · **freeze plug** a plug of frozen salt in the drain line · **redox** the oxidation-reduction state of the salt · **intergranular cracking** cracks along grain boundaries · **ASME Sec. III Div.5** the high-temperature nuclear construction code

Source This textbook §12.1 · salt boils above 1400 °C; operation at 650-700 °C · Hastelloy-N is not listed in ASME Sec. III Div.5

FLiBe - high heat capacity, low conductivity, exponential viscosity

- Volumetric heat capacity **4707 kJ/(m³·K)** - above water. But conductivity near 1.0 W/(m·K) is 1/64 of sodium - treat it as **an ordinary high-Pr liquid**, not sodium
- As it cools **pumping gets harder** - the wrong way for natural circulation. Chloride salts (fast MSR) still lack measured density, viscosity and conductivity - always cite the source



Density and viscosity of FLiBe (solid) and sodium (dashed) - the orange line is the 459 °C FLiBe melting point · textbook Fig. 12.1

VISCOSITY M

14.9 mPa·s

DENSITY P = 2245 - 0.424 T

2,033 kg/m³

VISCOSITY VS WATER AT 20 °C

×14.9

SALT TEMPERATURE

500 °C

FLiBe 2LiF-BeF₂ (66-34 mol%) - lithium enriched to over 99.99 % Li-7 to avoid Li-6 absorption · **FLiNaK** a LiF-NaF-KF salt · **eutectic** the lowest-melting composition

Source This textbook §12.2, Table 12.1, Fig. 12.1 · density measured 2022 (447-820 °C, 0.3 %) · viscosity $\mu = 1.16 \times 10^{-4} \exp(3755/T[K])$, $\pm 10-20\%$ · $k \approx 1.0$ ($\pm 10-50\%$) · $c_p \approx 2390$ ($\pm 2-3\%$) · viscosity values computed from the §16.3 formula

MSRE - the original, and the only measured data

- Validation data for every MSR code today comes **essentially from here** - including the world's first U-233 operation, 2 October 1968

ITEM	VALUE
Power	10 MWt design, about 7.4 MWt reached
Critical · U-233 · final shutdown	1 Jun 1965 (U-235 33 %) · 2 Oct 1968 · Dec 1969
Fuel salt	LiF-BeF ₂ -ZrF ₄ -UF ₄ 65-29.1-5-0.9 mol% · melts at 434 °C
Temperature · flow	632 → 654 °C (ΔT 22 °C) · 1200 gpm ≈ 170 kg/s
Loop transit	25.2 s (32 % outside the core)
Moderator · structure	graphite CGB, 3.05 × 1.016 cm channels · Hastelloy-N
Operating time	11,555 equivalent full-power h · fuel pump 19,405 h
Isothermal temperature coefficient (measured)	-13.14 ± 0.36 pcm/K

What MSRE **proved**, and what it left open

- Even with rods **deliberately withdrawn** at full power, fuel-salt expansion alone held the power down

Proved

- 1 over four years of circulating salt - no plugging, no pump failure
- 2 fuel changed in operation - U-235 → U-233
- 3 Xe stripping cut Xe-135 poisoning to about 1/6
- 4 a strong negative coefficient, -13.14 pcm/K
- 5 power held by salt expansion despite rod withdrawal

Left open

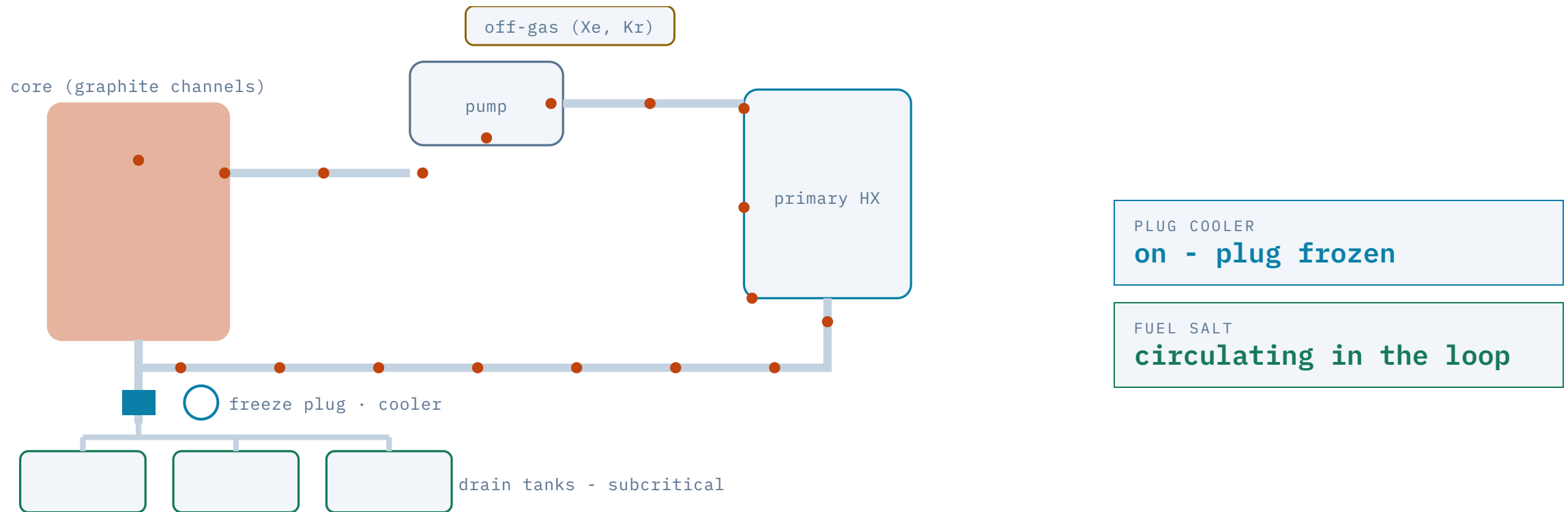
- 1 tellurium cracking in Hastelloy-N - Nb helped short-term
- 2 the tritium balance never closed
- 3 polymerised oil vapour clogged off-gas filters

Xe stripping sweeping xenon out with helium · **poisoning** absorbers like Xe-135 eating reactivity · **intergranular cracking** cracks along grain boundaries · **mass balance** the account of what goes in and out

Source This textbook §12.3 · 6-10 % of calculated tritium reached the containment atmosphere and another 6-10 % the heat-removal system - matching it needed wall permeability cut 1000-fold

The primary loop, **freeze plug**, drain tanks and off-gas

- The plug stays frozen only **while the cooler runs** - lose power, the cooler stops, the plug melts, and fuel salt drops by gravity **into tanks that cannot go critical**
- The off-gas system sweeps Xe and Kr into hold-up tanks - in ch. 15 it becomes **the main source term**



Normal operation

Loss of power

active cooler · **drain tank** subcritical storage below the core · **off-gas** the system sweeping out
hold-up tanks

Source This textbook §12.4, Fig. 12.2; §15.1 · drain tanks have a geometry that cannot go critical (several flat, wide tanks) · the drawing shows the sequence only (time arbitrary) - public quantitative data on freeze-plug melt time are hard to find (ch. 15)

Telling the names apart - liquid-fuel MSR and FHR

- Two quite different things hide under “MSR” - blur them in a talk and you will be corrected
- “Kairos is building a molten salt reactor” is, precisely, **“a salt-cooled high-temperature reactor”**

	① LIQUID-FUEL MSR	② FHR (SALT-COOLED HIGH-TEMPERATURE)
Fuel	dissolved in the salt, flowing	solid TRISO
Role of the salt	fuel and coolant	coolant only
Precursor drift (ch. 13)	yes	no
Closest relative	—	neutronically an HTGR, thermally an MSR
Examples	MSRE · IMSR · ThorCon · TMSR-LF1 · MCRE · MSFR	Kairos KP-FHR · Hermes

Why MSR again now - thorium, high temperature, the sea

- After MSRE ended in 1969 the MSR was **forgotten for forty years** - three reasons revived it, and the third matters most to our lab

1

Thorium

Breeding works best if Pa-233 (27-day half-life) is taken out and allowed to decay - impossible with solid fuel. TMSR-LF1 demonstrated Th-U conversion in Oct 2024.

2

High temperature

700 °C-class heat at atmospheric pressure - hydrogen, steel, petrochemicals.

3

The sea

Light low-pressure vessels, no phase change to slosh, and a freeze plug that works by gravity even with the ship heeled - a structural fit for floating plants (ch. 14).

Th-U cycle Th-232 capturing a neutron and becoming U-233 via Pa-233 · **Pa-233** the intermediate - left in the core it absorbs neutrons and spoils breeding

Source This textbook §12.6 · a forty-year gap after MSRE ended in 1969 · Pa-233 half-life 27 days · TMSR-LF1 confirmed Pa-233 after loading thorium in Oct 2024 and running 10 days at full power

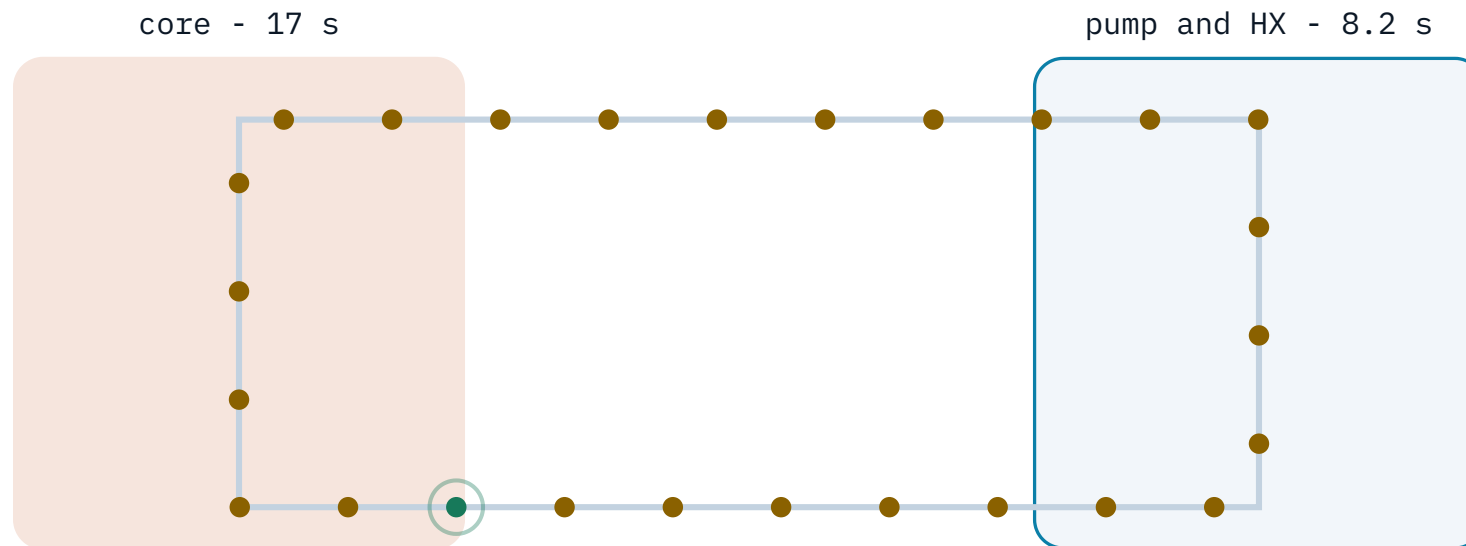
13

The precursors drift out

Reactors are controllable because of delayed neutrons. When the precursors that emit them flow out of the core with the fuel, the very scale of reactivity changes. It is the first physics to learn in MSR analysis.

Where delayed neutrons come from - β_{eff} in flowing fuel

- A precursor that decays while out in the pump and exchanger **contributes nothing to the core** - $\beta_{\text{eff, circulating}} < \beta_{\text{eff, static}}$
- At MSRE the loss is **34 %** - not a small correction but **a change in the reactivity scale itself**. A smaller β_{eff} makes the same reactivity more dangerous



DECAYS IN THE CORE
neutron counts

DECAYS OUTSIDE
lost

MSRE · U-235
654 → 431 pcm

yellow = precursor · green ring = decays in core (counts) · grey = decays outside (lost)

precursor a fission product that emits a delayed neutron on beta decay · β_{eff} effective delayed-neutron fraction · \$ (dollar) a reactivity equal to β_{eff} · **generation time** the mean time of one neutron generation

Source This textbook §13.1 · prompt neutrons within 10^{-4} s; about 0.65 % come seconds to tens of seconds later from precursor beta decay · β : U-235 about 0.65 %, Pu-239 about 0.21 %, U-233 about 0.27 % · the drawing is a concept - numbers on the following slides

The precursor transport equation - an advection term appears

- Because of this one term the precursor equation must be solved **over the whole loop, not just the core** - born in the core, carried along, decaying anywhere, some returning
- Short groups have $f \approx 0$ but mostly **decay inside the core**, losing about 2 %. Long groups spread round the loop and **lose the out-of-core share, $8.2/25.2 \approx 33\%$** - the longer the half-life, the bigger the loss
- Each group loses a different share, so **the shape of the delayed-neutron spectrum changes** - correcting a single β_{eff} is not accurate

solid fuel $\partial c_i / \partial t = (\beta_i / \Lambda) n - \lambda_i c_i$

liquid fuel $\partial c_i / \partial t + \nabla \cdot (c_i \mathbf{v}_f) = (\beta_i / \Lambda) n - \lambda_i c_i$

share of those leaving that return alive $f_i = e^{-\lambda_i \tau_{\text{out}}}$

MSRE RESIDENCE

$$17 + 8.2 = 25.2 \text{ s}$$

LONGEST GROUP (56 S) · LOST

$$\approx 33\% \cdot f \approx 0.90$$

SHORTEST GROUP (0.23 S) · LOST

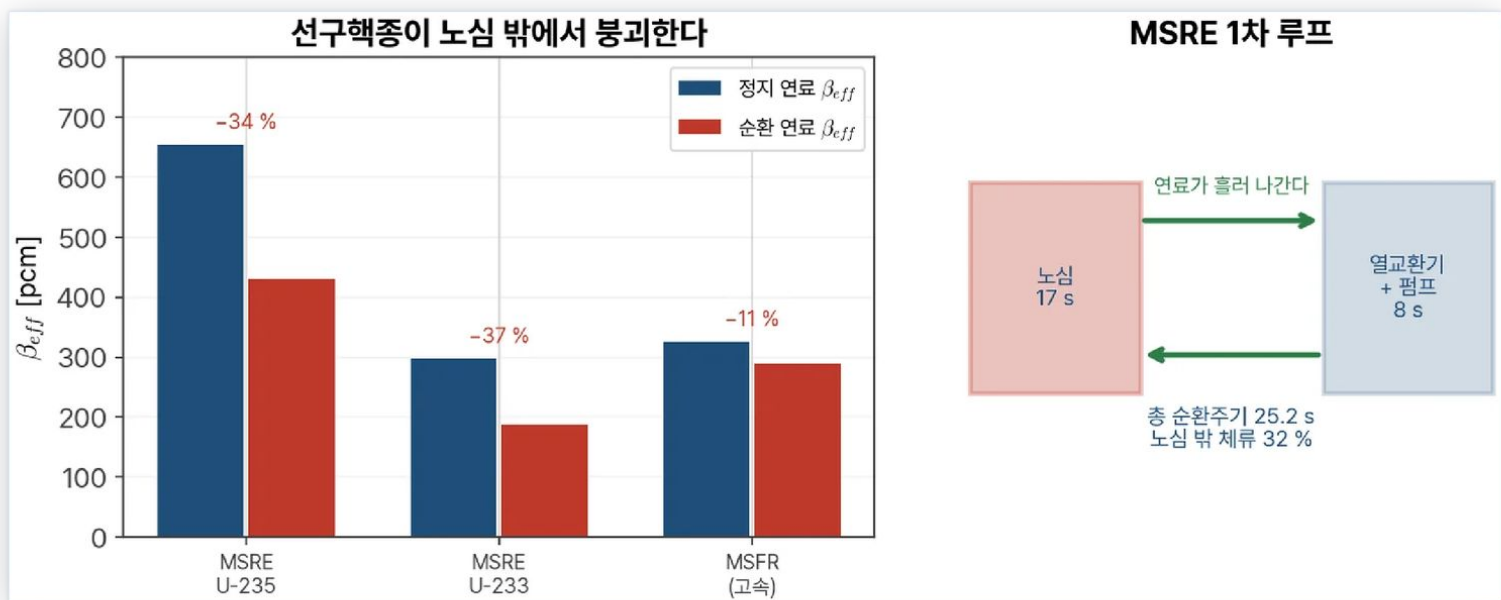
$$\approx 2\% \cdot f \approx 0$$

advection being carried along by the flow · **point kinetics** kinetics with one power value · λ_i decay constant of group i · Λ neutron generation time · $\mathbf{v}_f$ fuel-salt velocity field

Source This textbook §13.2 · MSRE loop transit 25.2 s, 17 s in the core, $\tau_{\text{out}} = 8.2 \text{ s}$ · group constants from $\lambda \approx 3.0 \text{ s}^{-1}$ (0.23 s half-life) to 0.0124 s^{-1} (56 s) · losses assume uniform production in the core and plug flow (revised §13.2)

β_{eff} in circulating fuel - calculated and measured

- That it was **measured** matters - stop the pump and precursors stay in the core, so reactivity rises. This start/stop test is **essentially the only measured validation for precursor-drift models**
- **Assuming uniform velocity can overstate circulating β_{eff} by up to 20-30 %** - do not trust a β_{eff} that does not state its simplification



REACTOR	STATIC	MOVING	LOSS
MSRE U-235	654	431	-34 %
MSRE U-233	298	188	-37 %
MSFR (fast)	325~407	285~299	-8~-26 %

MSRE CIRCULATION LOSS · U-235

212 ± 6 pcm measured · 223 calc.

β_{eff} reduction from precursor drift (left) and MSRE loop residence times (right) · textbook Fig. 13.1

MSFR the European molten-salt fast reactor concept (3000 MWt) · **pcm** 10^{-5} in reactivity · **saturation** the value barely moving as flow changes

Source This textbook §13.3, Table 13.1, Fig. 13.1 · measured by starting and stopping the MSRE fuel pump - 212 ± 6 pcm measured against 222 ± 10 pcm calculated · MSFR loop time 4 s, half the fuel in the core

Is **point kinetics** enough - five cases where it is not

- Mostly **point kinetics will do** - at most 3.5 pcm apart for pump start-up. Uniform transients (flow, power) are fine with it
- What the five exceptions share - **spatially non-uniform events**

1

Channel blockage

flow blocked locally

2

Fuel drain

geometry changes as
fuel drains

3

Gas entrainment

cover gas drawn in

4

Cold-slug injection

a cold slug enters
locally

5

Overfuelling

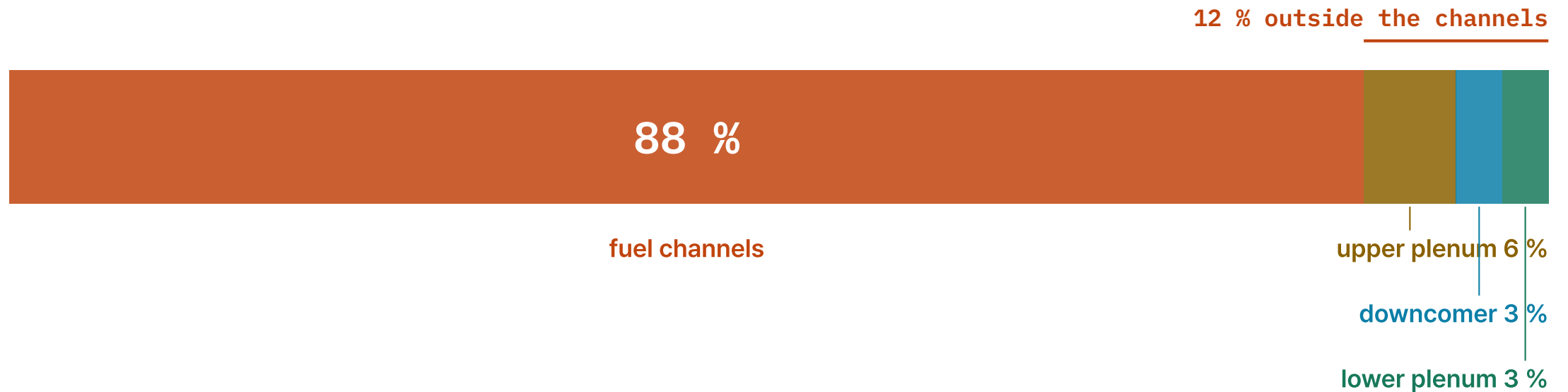
composition jumps

spatial kinetics kinetics that also resolves the flux shape · **Griffin, Squirrel** INL neutronics codes · **overfuelling** adding a lot of fuel at once so composition jumps

Source This textbook §13.4 · a 2025 Griffin/Squirrel comparison - at most 3.5 pcm between spatial and point kinetics for pump start-up

Decay heat flows too - 12 % of fission outside the core channels

- All the fission products flow, so decay heat sits in **pipes, exchangers and drain tanks** - drain the fuel and the heat goes with it → **the drain tanks need their own cooling**
- Some fission happens outside the core too - **assuming heat only in the core misplaces 12 % of it**



where fission occurs in the MSRE model

downcomer the down-flow path outside the core · **plenum** the open volume above or below the core · **drain-tank cooling** removing decay heat from drained salt

Source This textbook §13.5 · fission distribution in the MSRE model - fuel channels 88 %, upper plenum 6 %, downcomer 3 %, lower plenum 3 %

Online cleanup - gases, noble metals, rare earths and Pa

- Liquid fuel means **chemistry can be done while running** - Xe-135 poisoning down to about 1/6, so the LWR restart limit (xenon dead time) is much weaker
- ③ **has never been demonstrated** (not even at MSRE) - the MSR's biggest promise and least-tested part. A wide field for research

1

Gas removal

Xe and Kr are insoluble - sweep them out with helium. Xe-135 poisoning to about 1/6

shown at MSRE

2

Noble metals

Mo, Ru, Tc plate out on metal - a collector can be built for them

concept

3

Rare earths and Pa

a breeder must strip neutron-eating rare earths and let Pa-233 decay apart

never demonstrated

xenon dead time the post-shutdown period when Xe-135 prevents restart · **noble metals** Mo, Ru, Tc and others that plate out on metal · **MSBR** ORNL's Molten Salt Breeder Reactor concept

Source This textbook §13.6 · MSRE sprayed 1.5-5 L of helium a minute (2.4 nominal) in the pump bowl, recirculating about 4 % of pump flow · the MSBR concept replaced graphite every four years and ran a chemistry loop continuously

14 MSRs being built, and the sea

An exact picture as of September 2026. This chapter ages fastest. Company names and specs change often, so set the classification axes first and date every number.

The four axes that separate MSR projects

- Kairos KP-FHR is **solid fuel · fluoride · thermal · HALEU**; TerraPower MCFR is **liquid fuel · chloride · fast · HALEU** - both “MSRs”, entirely different reactors

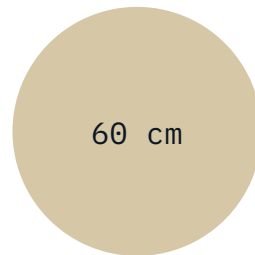
AXIS	BRANCH ①	BRANCH ②	KAIROS KP-FHR	TERRAPOWER MCFR
Fuel form	liquid (dissolved)	solid FHR (TRISO + salt)	②	①
Salt	fluoride (FLiBe etc.)	chloride (NaCl- <chem>UCl3</chem> etc.)	①	②
Spectrum	thermal (graphite)	fast (no moderator)	①	②
Fuel cycle	uranium (LEU/HALEU)	thorium	①	①

Why low pressure still gives high power - **heat capacity and one pipe**

- FLiBe's ρc_p of **4707 kJ/(m³·K)** beats water's 3870 at 15.5 MPa and 310 °C - more heat per volume moved
- **One low-pressure pipe carries 400 MW** - a completely different fabrication problem from 15 MPa piping

$$Q = \rho \dot{V} c_p \Delta T \quad \rightarrow \quad \dot{V} = Q / (\rho c_p \Delta T)$$

low-pressure pipe section (scale - 1 cm = 2 px)



VOLUME FLOW $\dot{V}$

0.85 m³/s

PIPE BORE (3 M/S)

60 cm

THERMAL POWER Q

400 MWt

CORE ΔT

100 K

volumetric heat capacity heat stored per volume per kelvin · $\dot{V}$ volume flow · **IMSR** Terrestrial Energy's Integral Molten Salt Reactor

Calculation This textbook §14.2 · FLiBe at 650 °C $\rho c_p = 1969 \times 2390 = 4.71 \times 10^6$ J/(m³·K) · $\dot{V} = Q/(\rho c_p \Delta T)$ · pipe velocity 3 m/s · early IMSR 400 MWt with ΔT 100 K → 0.85 m³/s, 0.28 m², about 60 cm bore

Kairos Power **KP-FHR** - the solid-fuel FHR, and Hermes 1 and 2

- **Why Kairos chose 316H is instructive** - Hastelloy-N performs better but is not in ASME Div.5. **Code listing, not performance, chose the material**, and the 650 °C limit too

ITEM	VALUE · STATUS
Type	FLiBe coolant + annular TRISO pebbles (not fuel salt) · online refuelling
Conditions	650 °C outlet · near atmospheric · HALEU 19.75 % · 316H · ET-10
Hermes 1 · 35 MWt	non-power demonstration. NRC construction permit Dec 2023 - the first US non-LWR permit. Safety-related work 1 May 2025. Deadline moved to 30 April 2029
Hermes 2 · 50 MWe	permit Nov 2024 · raised to 50 MWe after the Google-TVA PPA · nuclear construction 17 April 2026 · grid target 2030
Commercial	2 × 75 MWe = 150 MWe, up to 450-600 MWe

FHR fluoride-salt-cooled high-temperature reactor · **316H** high-temperature grade 316 stainless · **ET-10** an Ibiden graphite grade · **PPA** power purchase agreement · **TVA** Tennessee Valley Authority

Source This textbook §14.3 · **as of September 2026** · Hermes 1 completion moved from end-2026 to 30 April 2029 (28 months, NRC, 21 April 2026) · the only ASME Sec. III Div.5 materials are 2¼Cr-1Mo, 9Cr-1Mo, 304SS, 316SS and Alloy 800H

IMSR and ThorCon - replaceable core units and shipyard construction

- Both solve component life by **swapping the whole unit rather than maintaining it** - every 7 years for IMSR, 8 for ThorCon

Terrestrial Energy — IMSR

Canada/US · integral liquid-fuel

- core, pump and HX in one unit - replaced every 7 years
- was 400 MWt / 195 MWe → now 882 MWt / 392 MWe · 585 °C peak
- NRC PDC approval late 2025 · no construction application
- June 2026 Texas A&M RELLIS ground lease, ~77 acres · Nasdaq listing Oct 2025

ThorCon

US/Indonesia · shipyard-built

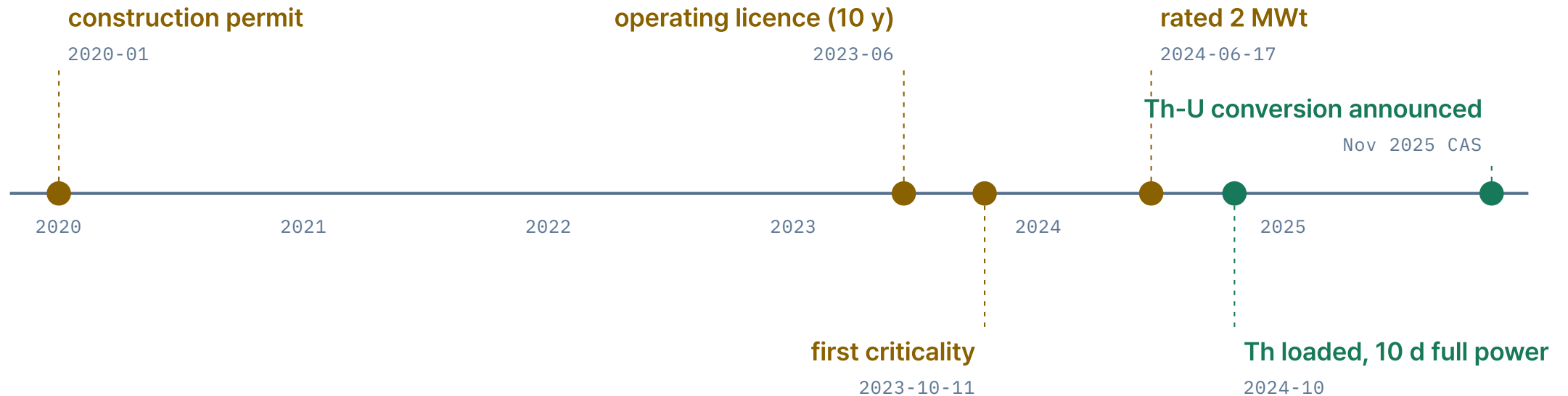
- 500 MWe = two 250 MWe Cans - built in blocks, towed on a barge
- each Can is swapped after 8 years
- Feb 2025 site programme to BAPETEN - Indonesia's first nuclear application
- Aug 2025 Kelasa island site approved · operation before 2032 · no permit yet

PDC principal design criteria · BAPETEN Indonesia's nuclear regulator · Can ThorCon's replaceable reactor module

Source This textbook §14.3 · as of September 2026 · IMSR principal design criteria approved by NRC late 2025 · ThorCon filed Indonesia's first nuclear licence application with BAPETEN

TMSR-LF1 - the only liquid-fuel MSR running today

- The world's only **operating liquid-fuel MSR** - after loading thorium in Oct 2024, ten days at full power **detected Pa-233 and demonstrated Th-U conversion**



TMSR-LF1 Thorium Molten Salt Reactor - Liquid Fuel 1 · **Th-U conversion** Th-232 becoming U-233 · **Pa-233** the intermediate - finding it proves conversion happened

Source This textbook §14.3, Table 14.1 · Wuwei, Gansu · 2 MWt · FLiBe + ZrF₄ + UF₄ (HALEU 19.75 %) + ThF₄ · 630 → 650 °C · graphite-moderated · follow-ons TMSR-LF2 (10 MWt) and TMSR-SF2 (100 MWt) unscheduled

MCRE, Natura MSR-1 and other MSRs

- MSR-1 is a **case of a changed licensing path** - it held the first US liquid-fuel MSR permit (NRC) yet turned to the DOE pilot route to reach criticality first

DESIGN	SIZE · TYPE	STATUS (SEPT 2026)
TerraPower / Southern — MCRE	200 kWt · chloride fuel salt · INL ZPPR cell	to be the first fast-spectrum circulating-fuel reactor. First enriched chloride fuel-salt batch made late Sept 2025 - 72-75 batches to criticality, operation expected 2028. Partners include the marine-nuclear firm Core Power
Natura Resources — MSR-1	about 1 MWt research reactor · ACU campus	NRC permit Sept 2024 (first US liquid-fuel MSR). OTA with DOE Dec 2025 - criticality first under the DOE pilot. Whether it went critical needs checking
Copenhagen Atomics	100 MWth · unpressurised D ₂ O moderator · thorium blanket	agreement with Switzerland's PSI; thorium molten-salt critical experiment targeted for 2026
Moltex → Nuclea Energy	SSR-W 300-500 MWe fast · static-salt fuel pins	Nuclea bought the whole Moltex portfolio in Aug 2026, WATSS recycling included - read "Moltex" as "former Moltex, now Nuclea"

MSRs at sea - **Saltfoss CMSR, MARINA** and a salt-reactor car carrier

- In the first half of 2026 alone **two Korean-led marine MSR designs** won class approval - the setting for THINK Lab's floating-MSR project
- **"NaOH-moderated CMSR" in the present tense is wrong** - moving to LEU over HALEU supply risk in early 2023 switched it to graphite. Say "was originally NaOH-moderated"

Saltfoss CMSR

Denmark · power barge

- 100 MWe units · 2 per barge (200 MWe)
- up to 8 (800 MWe)
- atmospheric · 12-year fuel cycle
- ABS feasibility statement - NTQ stage 1 of 5
- first barge targeted for H1 2030

MARINA

KAERI · KRISO · Samsung Heavy

- ABS Approval in Principle, 16 July 2026
- 15,000 TEU container ship · 25 knots
- two reactors · Panama-canal hull
- detailed specs undisclosed

Salt-reactor car carrier

HD HHI · HD KSOE · Hyundai Glovis · G-Marine · KAERI

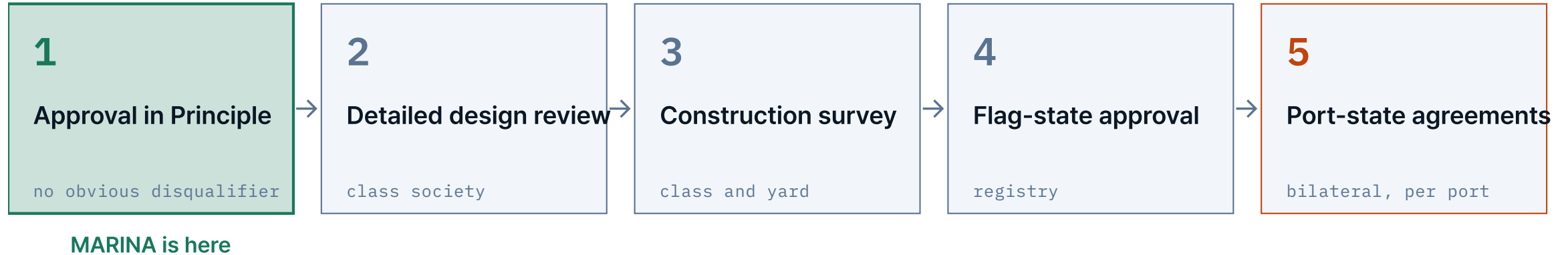
- Lloyd's Register design approval, June 2026
- a 7,000 CEU car carrier
- the second Korean-led marine MSR approval

CMSR Compact MSR · **ABS, LR** American Bureau of Shipping and Lloyd's Register · **AiP** Approval in Principle · **NTQ** ABS New Technology Qualification (five stages) · **TEU, CEU** container and car units · **KRISO** Korea Research Institute of Ships and Ocean Engineering

Source This textbook §14.4 · as of **September 2026** · Seaborg Technologies → Saltfoss Energy (Apr 2025) · MARINA's detailed specs (power, fuel salt, temperatures) undisclosed

Maritime rules - the IMO code and where AiP sits

- The basis is **the 1981 IMO code** - no real revision in 45 years, nothing on MSRs. The biggest practical barrier is **port-state acceptance** (a bilateral deal per port, negotiated years ahead)
- **Do not read "AiP granted" as "licensed"** - the class society found "no obvious disqualifier", stage 1-2 of 5 in NTQ terms



A gap in liability

The 1962 Brussels convention never entered into force - insurance is negotiated case by case. The IMO revision started at SDC in Jan 2026, no end date

IMO International Maritime Organization · **MSC** Maritime Safety Committee · **SDC** Ship Design and Construction sub-committee · **SOLAS** Safety of Life at Sea convention · **flag state, port state** the ship's registry, and the country of the port it enters

Source This textbook §14.4 · IMO Resolution A.491(XII), 1981 Code of Safety for Nuclear Merchant Ships · full revision decided at MSC 110, 26 June 2025 · SDC sub-committee from its January 2026 session · the 1962 Brussels convention on nuclear ships never entered into force

Stale information still widely quoted - checked in September 2026

- MSR companies and specs change often - six cases turned up just while writing this chapter

OLD INFORMATION

NOW (SEPT 2026)

Seaborg Technologies

Saltfoss Energy (renamed Apr 2025)

Seaborg CMSR = NaOH-moderated

graphite-moderated (changed 2023)

Moltex Energy

Nuclea Energy (acquired Aug 2026)

IMSR 400 MWt / 195 MWe

882 MWt / 392 MWe (redesigned)

ThorCon 500 MWt

500 MWe (the unit error is widespread)

USNC MMR 15 MWt

NANO Nuclear KRONOS, docketed at 45 MWth

ARIS the IAEA Advanced Reactors Information System · MWt, MWe thermal and electrical output - mixing them is the most common error

Source This textbook §14.5, Table 14.2 · check via ⓘ IAEA ARIS design reports ⓘ the NRC application pages ⓘ company announcements - quoting only the press copies these errors

Why at sea - **four things that fit**, three that get harder

- **Freezing is worse on a ship** - every long lay-up, blackout or dry-docking means holding the whole plant above 450 °C

What fits

- 1 low pressure - no 15 MPa piping on a moving deck
- 2 no phase change - no sloshing-boiling coupling (free surface remains)
- 3 gravity shutdown - still works heeled
- 4 high power density, small core - fits a hull

What gets harder

- 1 freezing (459 °C) - above 450 °C at every lay-up, blackout, docking
- 2 maintenance - a hot primary, far harder at sea
- 3 source term - off-gas and tritium (ch. 15)

sloshing a free surface rocking in its tank · **dry-docking** lifting a ship out of the water · **source term** the amount and kind of radioactivity that could reach the environment

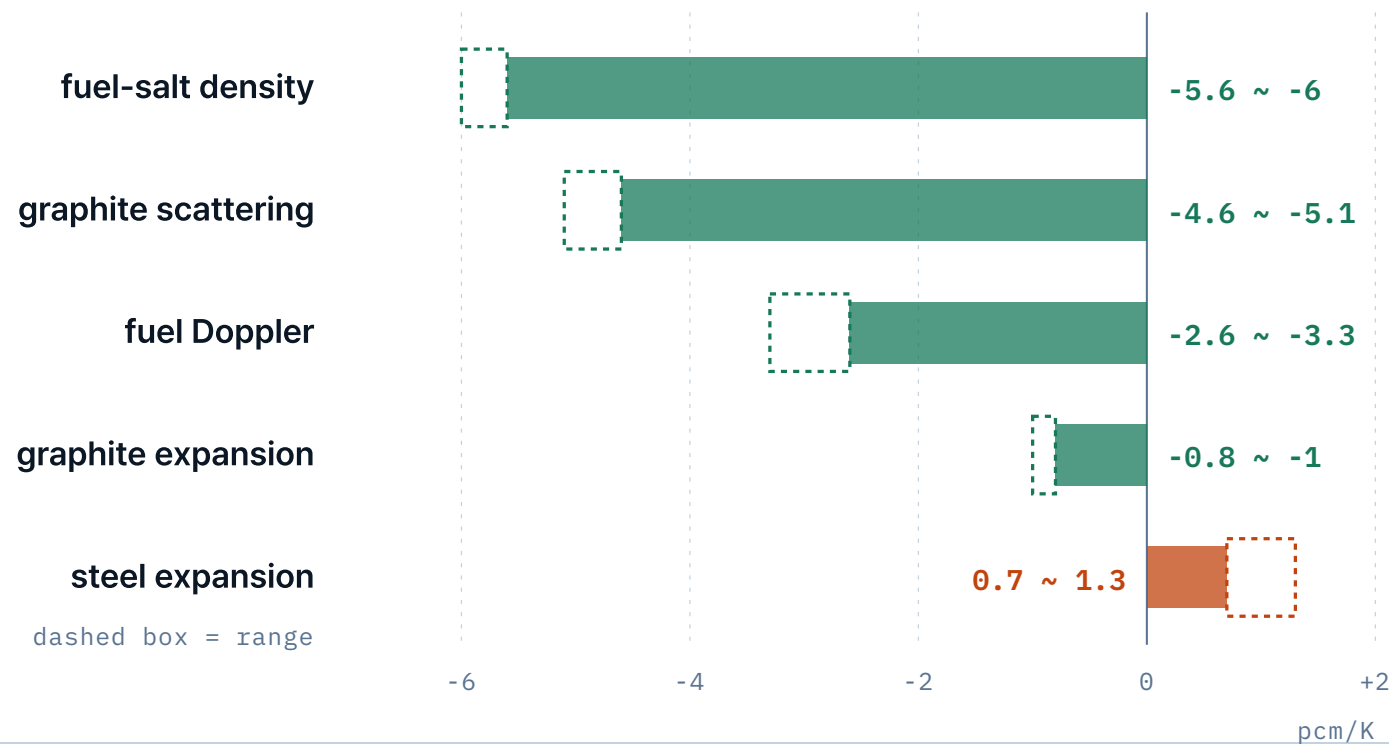
Source This textbook §14.6 · drain-tank criticality with the ship heeled is a topic no one has properly addressed yet

15 Freeze plugs and off-gas

The MSR safety case is strong. So are the open problems. Containment moves from the building to the off-gas system, and a chemistry probe guards structural integrity.

Strong arguments, and the MSRE temperature coefficient

- The biggest term is **fuel-salt expansion** - hot fuel expands out of the core. **The fuel itself is a safety system**
- But **public quantitative data on freeze-plug melt time and reliability are scarce** - cite Tiberger et al. (Nucl. Eng. Des., 2020) and ORNL-TM-0128 directly



MSRE MEASURED TOTAL

-13.14 pcm/K

THE OTHER THREE

② low pressure - over 700 K to boiling

③ no cladding - no failure, oxidation or hydrogen

④ freeze plug + drain tanks

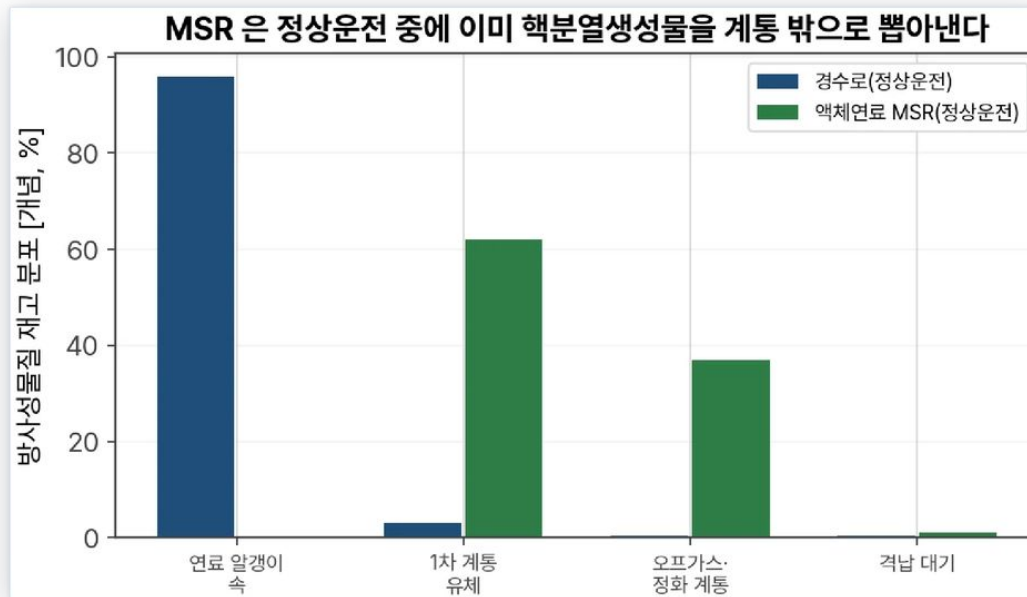
isothermal coefficient reactivity change per kelvin of uniform heating · **graphite thermal scattering** the spectrum shift as graphite heats ·

Serpent2 a Monte Carlo neutron transport code

Source This textbook §15.1, Table 15.1 · MSRE measured isothermal coefficient -13.14 ± 0.36 pcm/K · contributions recomputed with Serpent2 · with rods deliberately withdrawn at full power, output limited itself near 9 MWt with no operator action

MSR containment - the off-gas system, not the building

- The first factor of an LWR source term (fuel → system) means nothing here - fission products are in the system from the start, and volatiles are **already in the off-gas during normal operation**
- **"MSRs have a small source term" is half true** - the off-gas is a standing high-concentration point, and it is not built like a reactor building. Licensing will dig here hardest



$$\text{release} = I_0 \times f_{\text{fuel} \rightarrow \text{sys}} \times f_{\text{sys} \rightarrow \text{cont}} \times f_{\text{cont} \rightarrow \text{env}}$$

MSBR OFF-GAS DECAY HEAT

21.3 MW · 1 % of output

GAS HEAT LOAD - WITHIN 1 H · AFTER 2 D

80 % · < 1 %

Where the radioactive inventory sits in normal operation (concept) - inside the fuel for an LWR, in the fluid and off-gas for an MSR · textbook Fig. 15.1

source term radioactivity that could reach the environment · **hold-up tank** a tank that holds gas while it decays · **aerosol, mist** fine solid or liquid particles carried in gas

Source This textbook §15.2, Fig. 15.1 · for the MSBR (2250 MWt) concept - off-gas decay heat 21.3 MW (about 1 % of output), about 9 % of it as aerosol and mist, 80 % of the gas heat load within 1 h, under 1 % after 2 days · 2 h hold-up plus 47 h decay · 90 days of hold-up is 17 Xe-133 half-lives

Unquantified - high-importance, low-knowledge items in the ORNL PIRT

- Honestly, the quantitative basis of MSR safety is **still thin** - salt-particle estimates spread over **four orders of magnitude (10^{-8} to 10^{-4} g/cm³)**, not licensing precision

1 Droplet aerosolisation on a salt spill

how far spilled salt travels

2 Salt-substrate interaction and corrosion

how fast structures corrode

3 Radionuclide volatilisation from salt

what leaves, and how much, with temperature

4 Speciation and flowing salt with oxygen

in what form, and how it reacts with air

5 Salt / cover-gas partitioning

the thermochemical data are missing - PIRT top priority

PIRT Phenomena Identification and Ranking Table · **speciation** which chemical form an element takes · **volatilisation** passing from liquid to gas
· **daughter** a nuclide produced by decay

Source This textbook §15.3 · under 1 % of MSRE salt became aerosol, carrying noble metals and Kr/Xe daughters (Cs, Ce, Ba, Rb, Sr, Y) · salt-particle concentration estimates span 10^{-8} to 10^{-4} g/cm³

Tritium - the MSR's chronic condition

- Tritium **walks straight through hot metal**. At MSRE 6-10 % reached the containment atmosphere and another 6-10 % the heat-removal loop - **matching it needed permeability cut 1000-fold** (model uncertainty)
- It matters more at sea - on land it goes up a stack, but a ship's **release point keeps moving and every port has its own rules**

PLANT	TRITIUM PRODUCED OR RELEASED
410 MWe pebble-bed FHR (equilibrium)	93 TBq/d (2,500 Ci/d) · 666 TBq/d at first
FHR in general	500-5,000 Ci/d
2250 MWt MSR	about 89.5 TBq/d ≈ 250 mg/d

FOUR MITIGATIONS

- ceramic permeation barriers ·
- sweep gas and membrane stripping ·
- graphite and other getters ·
- hydrogen counter-diffusion

Materials and chemistry as **safety systems** - redox, Hastelloy-N

- **One chemistry probe guards structural integrity** - a coupling far beyond boron control in an LWR. System codes lack it, and no one has built the chemistry model properly
- Hastelloy-N is **not listed** in ASME Sec. III Div.5 - the real bottleneck is **code listing, not technology**

U(IV)/U(III) RATIO	TEMPERATURE	RESULT
below 30-40	up to 760 °C	chromium corrosion acceptable
42	760 °C	no tellurium cracking
60	760 °C	severe intergranular cracking
85	800 °C	cracks up to 148 µm deep

HASTELLOY-N DEGRADATION

- ① **helium embrittlement - Ni-58**
helium cavities at grain boundaries
- ② **tellurium cracking - redox-dependent**
- ③ **weak long-term strength above 650-700 °C**

U(IV)/U(III) ratio tetravalent to trivalent uranium in the salt - the redox indicator · **intergranular cracking** cracks along grain boundaries · **helium embrittlement** irradiation helium forming cavities at grain boundaries

Source This textbook §15.5 · corrosion at 800 °C / ratio 85 is more than twice that at 760 °C / ratio 42 · surface chromium 3.5 wt% at low redox, 1.7 wt% at high · Hastelloy-N is allowed only to 704 °C under ASME Sec. VIII Div.1

Defence in depth redrawn - LWR against MSR

- The most dramatic change - **level 4 disappears**. Half of severe-accident research (core melt, MCCI, hydrogen, in-vessel retention) no longer applies; the fuel is already liquid
- Instead **level 1 becomes safety-grade** - chemistry failure is chronic, over years, and fits PSA badly. How to put “loss of redox control” into an event tree is a good research topic

LEVEL	LWR	MSR
1 Normal operation	materials and chemistry secondary	chemistry (redox) and freeze protection are safety functions
2 Anomaly detection	pressure and level	temperature, flow, chemistry
3 Accident mitigation	ECCS, containment spray	freeze plug + drain tanks (passive)
4 Severe accidents	core-melt management	fuel already liquid - no “melt”
5 Emergency response	evacuation planning	off-gas and tritium release

PSA probabilistic safety assessment · **event tree, fault tree** accident paths and failure combinations drawn as trees · **chronic degradation** damage building slowly over years

Source This textbook §15.6, Table 15.3 · the five-level frame is from *SMR Design Requirements and ESF* · event and fault trees from *Introduction to PSA*

16

Running a system code on molten salt

Helium and sodium mostly run in the Volume III code once properties and correlations change. Molten salt does not. It is the only case that needs the code structure itself changed - though less work than you might think.

Five things to change for molten salt

- The first two are the same kind of work as before; **the last three force the code to be restructured**

WHAT	WHY	KIND
Property class	salt tables - conductivity uncertain by 10-50 %	swap
Heat transfer · friction	validated for water - reassess for salt	swap
Where heat is made	in the fluid, not a fuel rod	restructure
Precursor transport	advection added to point kinetics (ch. 13)	restructure
Freezing	solidification and plugging below T_{melt}	restructure

property class the code object returning ρ , T and more from p and h · **closures** friction and heat-transfer correlations · **plugging** a flow path blocked by solidification

Source This textbook §16.1 · the RELAP5-3D documentation itself says heat-transfer and friction correlations were validated mainly for water and need separate assessment for molten salt

Changes 1 and 2 - heat in the fluid, and precursor transport

- In Volume III heat passes from a HeatStructure to the fluid - an MSR puts a **volumetric source straight into the energy equation**, with q_{fission} and q_{decay} fields per volume
- Precursors move **exactly as enthalpy was donor-celled in Volume III** - six scalars (six groups) on each Volume, `Junction.donor()` reused. Less work than it looks

Change 1 · heat in the fluid

$$\partial(\rho h)/\partial t + \nabla \cdot (\rho \mathbf{v} h) = q'''_{\text{fission}} + q'''_{\text{decay}} + \nabla \cdot (k \nabla T)$$

Change 2 · precursor group i in volume j

$$V_j \, dc_{i,j}/dt = \sum_{\text{in}} \dot{V}_k c_{i,k} - \sum_{\text{out}} \dot{V}_k c_{i,j} + V_j (\beta_i/\Lambda) n_j - \lambda_i c_{i,j} V_j$$

$$dn/dt = ((\rho - \beta)/\Lambda) n + \sum_i \lambda_i c_{i,\text{core}}$$

q''' heat per unit volume · **donor cell** first-order upwinding · `Junction.donor()` the Volume III upwind selector · **Volume** the control-volume class

Source This textbook §16.2 · at MSRE 12 % of fission is outside the core channels (upper plenum 6, downcomer 3, lower plenum 3 %) · only c_i inside core volumes enters the last term of the kinetics equation

Change 3 - freezing, and the effective heat-capacity method

- The simplest change and the easiest to miss - handle latent heat and the mushy zone, and make the Junction **lose area and gain friction** as it freezes
- Most system codes lack freezing and remelting, so many analyses **stop at "the moment it drops below the melting point"** - useful, but plug timing and plug propagation need the model

$$T_j < T_{\text{melt}} \Rightarrow \text{freezing starts}$$

$$c_{p,\text{eff}} = c_p + L / \Delta T_{\text{mushy}} \quad (T_{\text{sol}} < T < T_{\text{liq}})$$

$$A_{\text{jct}}(\varphi_s) , K_{\text{jct}}(\varphi_s)$$

FLIBE MELTING POINT

459 °C

MSRE FUEL SALT MELTS

434 °C

MATTERS MORE AT SEA

**long lay-ups, blackouts and
docking come far more often**

effective heat capacity spreading latent heat over the melting range as extra c_p · **mushy zone** the temperature range where solid and liquid coexist · **solid fraction** the frozen share of a volume

Source This textbook §16.2 · L latent heat of fusion, ΔT_{mushy} the solid-liquid range · once freezing starts, Junction area and loss coefficient become functions of solid fraction

The salt property class - density fits, **viscosity must follow temperature**

- Density agrees with measurement to **0.3 %**. Viscosity is the problem - fixed at 5.6 mPa·s, really 6.8 at 650 °C and **15.0 at 500 °C**. More than double over 150 K
- Near freezing it **must be a function of temperature** - and MSRE fuel salt differs again; cite ANL's SAM values

```
class MoltenSalt(DenseLiquid):
    def viscosity(self, T):    # T [K]
        return self.mu_A*np.exp(self.mu_B/max(T, self.T_melt))
    def frozen(self, T):
        return T < self.T_melt

flibe = MoltenSalt(2413.0, 0.488, cp=2390.0, k=1.1,
                  mu_A=1.16e-4, mu_B=3755.0, T_melt=732.15)
```

P(650 °C) · VOL. III VS MEASURED

1963 / 1969

M(650 °C)

6.8 mPa·s

M(500 °C)

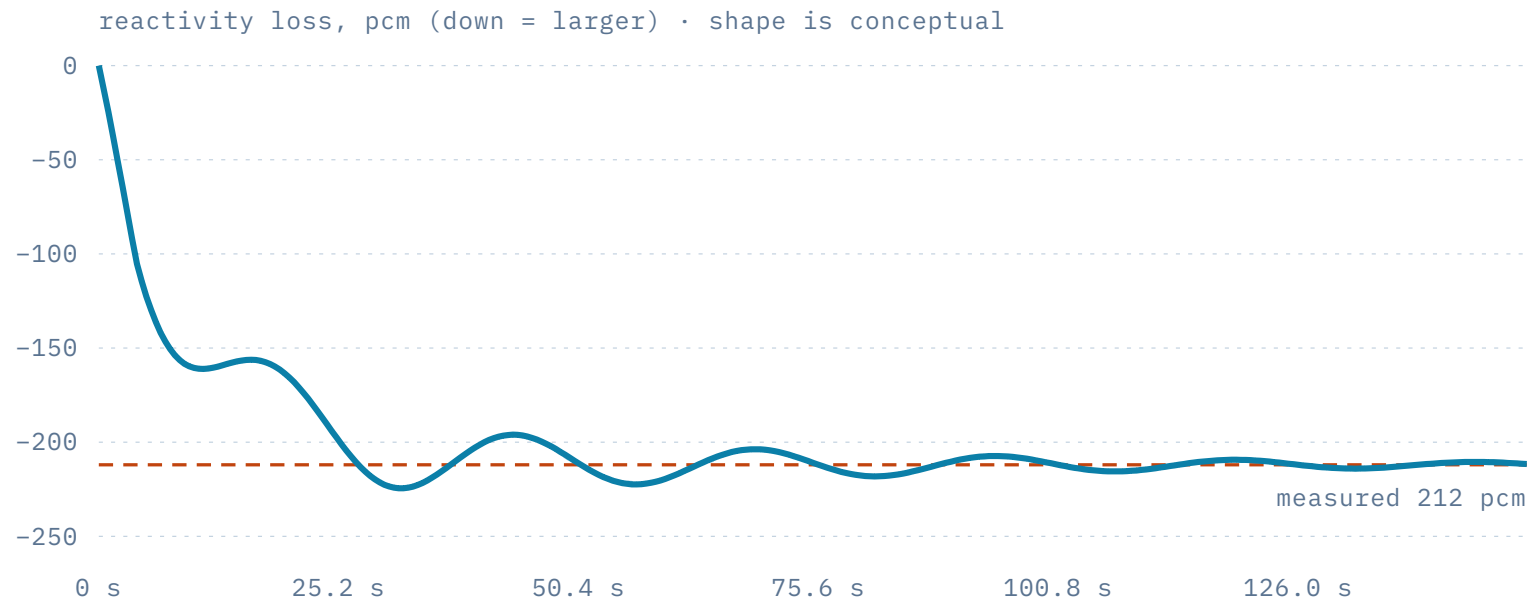
14.9 mPa·s

DenseLiquid Volume III's linear-density liquid class · **MoltenSalt** the subclass built here - adding temperature-dependent viscosity and a freeze check

Calculation This textbook §16.3 · Volume III FLiBe() = DenseLiquid(2413, 0.488, T in kelvin) · at 650 °C $2413 - 0.488 \times 923 = 1963$ against 1969 measured (0.3 %) · $\mu = 1.16 \times 10^{-4} \exp(3755/T)$ · MSRE salt (ANL SAM): $\rho = 2810.7 - 0.6707(T - 722.15)$, $\mu = 8.4 \times 10^{-4} \exp(4340/T)$, k 1.0, c_p 1884 · graphite ρ 1869.6, k 81.3, c_p 1698.4 (900 K)

The MSRE pump start-up benchmark - first test of any new MSR code

- The **only measured data** aimed directly at precursor drift - start the pump on a critical core, rods fixed, and compute the reactivity loss
- Reactivity ripples **every 25.2 s** as the first slug comes round - **see the ripple and transport is in right**; miss it and something has been averaged away



QUASI-STEADY LOSS (MEASURED)

212 pcm

RIPPLE PERIOD = TRANSIT TIME

25.2 s

BEFF STATIC → MOVING

654 → 431 pcm

transit time one trip round the loop · **quasi-steady** nearly steady once oscillation dies out · **damped oscillation** a swing that shrinks each cycle

Source This textbook §16.4 · inputs - 17 s in core, 8.2 s outside, 25.2 s loop, six groups (U-235), β_{eff} static 654 pcm, coefficient -13.14 pcm/K · expected - quasi-steady loss 212 pcm (measured), β_{eff} circulating 431 pcm, a decaying oscillation with a 25.2 s period · the curve is a concept shape

MSR system codes - and Korea's MARS

- **MARS-MSR is the code to watch here** - a Korean code applied to MSRs by Korean authors, reproducing the 25.5 s reactivity ripple. First candidate for the floating-MSR project

CODE	WHAT IT HAS	CAVEAT
SAM (ANL)	scalar transport for precursors, coupled to Griffin. MSRE model public (ANL/NSE-23/8)	radial velocity suppressed artificially to mimic the stringers (documented)
Griffin / Squirrel (INL)	spatial and point kinetics compared - the source of ch. 13's β_{eff} values	—
RELAP5 / TRACE	several adaptations	correlations are water-based - not to be used as is
MARS (KAERI / Pusan) ★	precursor transport · salt properties (LiF-BeF ₂ -ZrF ₄ -UF ₄ , LiF-BeF ₂ , KCl-UCl ₃ , FUNaK) · transported decay heat · MSR feedback. Benchmarked on MSRE pump start/stop and three natural-circulation tests	reproduced the ~25.5 s reactivity ripple

MARS KAERI's system thermal-hydraulics code (RELAP5 and COBRA lineage) · FUNaK a fluoride fuel salt · graphite stringer the graphite bars forming the core · multiphysics coupling solving neutronics and thermal hydraulics together

Source This textbook §16.5 · SAM's MSRE model is public as ANL/NSE-23/8 · MARS-MSR: J. Jeong, Y. Cho, H. Lee, B. Yun, Nucl. Eng. Technol. 58(10), 2026

Four empty seats in MSR thermal hydraulics

- All four need experiment and analysis together, and **our lab's tools - visualisation, scaling, CFD, system codes - reach them**. Empty means there is room to contribute

1 Freeze and remelt models

no system code has one properly - even more important at sea

2 Coupling to hull motion

natural circulation, cover-gas sloshing and drain-tank criticality under 6-DOF motion - essentially no literature. The R134a EDV tests and scaling apply

3 Chemistry-TH coupling

bringing redox-driven corrosion and long-term integrity into the model

4 Source-term models

salt / cover-gas partition data missing - PIRT top priority

six degrees of freedom a hull's three translations and three rotations · **EDV** emergency depressurisation valve · **R134a** a refrigerant used in similarity tests · **PIRT** Phenomena Identification and Ranking Table

Source This textbook §16.6 · the lab's R134a EDV dynamic-load tests, scaling methodology, flow visualisation, CFD and system codes

17

All three on one ruler

VHTR, SFR and MSR folded into one table. “Which reactor is better?” has no answer - it gets one only once the use and the yardstick are fixed.

Six axes for comparing reactors

- Fix the axes first and each reactor's place becomes clear - each axis has a different leader (Table 17.3 later)

1**Thermodynamics**

how hot, and at how low a pressure

2**Safety logic**

what stops an accident

3**Fuel cycle**

what it burns and what it leaves

4**Materials · chemistry**

what sets its lifetime

5**Maturity**

has it ever actually run

6**Analysis**

which code, validated on which data

Coolant properties on one page - **four cells** that stand out

- Each reactor's own hard problem comes from **one cell** of this table - not to memorise, but to read which cell stands out

PROPERTY	WATER	HELIUM	SODIUM	FLIBE
Density kg/m ³	727	3.29	835	1,969
c _p J/(kg·K)	5,320	5,195	1,264	2,390
ρc _p kJ/(m ³ ·K)	3,870	17	1,055	4,707
Conductivity W/(m·K)	0.54	0.37	64.2	1.0
Viscosity mPa·s	0.086	0.047	0.236	6.8
Prandtl number	0.86	0.66	0.0046	16
Melting point °C	0	—	97.8	459
Boiling point °C	345 (sat.)	—	883	>1400
Margin to boiling K	20	—	373	>750

① ρc_p

② Pr

③ boiling margin

④ melting

HEAT PER VOLUME - ONLY HELIUM IS TWO
ORDERS DOWN

**FLiBe > water > sodium >>
helium. So only the HTGR
needs 7 MPa**

volumetric heat capacity heat to raise 1 m³ by 1 K · **Prandtl number Pr** momentum diffusion ÷ heat diffusion · **boiling margin** temperature left between operation and boiling

Source This textbook, Table 17.1 · water 15.5 MPa 310 °C · helium 7 MPa 750 °C · sodium 0.1 MPa 500 °C · FLiBe 0.1 MPa 650 °C · volumetric heat capacity = ρc_p

The three compared (1 of 2) - design parameters

- **Power density differs by two orders** - the graphite VHTR runs 3-6 MW/m³, the unmoderated SFR 230-450. Same power, a much bigger VHTR core

ITEM	VHTR (HTGR)	SFR	MSR (LIQUID FUEL)
Coolant	helium	sodium	fluoride or chloride salt
Moderator	graphite	none (fast)	graphite or none
Fuel	TRISO particles	metal or oxide pins	dissolved in salt
Primary pressure	6~7 MPa	~0.1 MPa	~0.1 MPa
Outlet temperature	750~950 °C	500~550 °C	650~700 °C
Power density MW/m ³	3~6	230~450	3 (thermal) - 300 (fast)
Intermediate loop	none	yes	yes (secondary salt)
Efficiency	~40 %	~40 %	~45 %

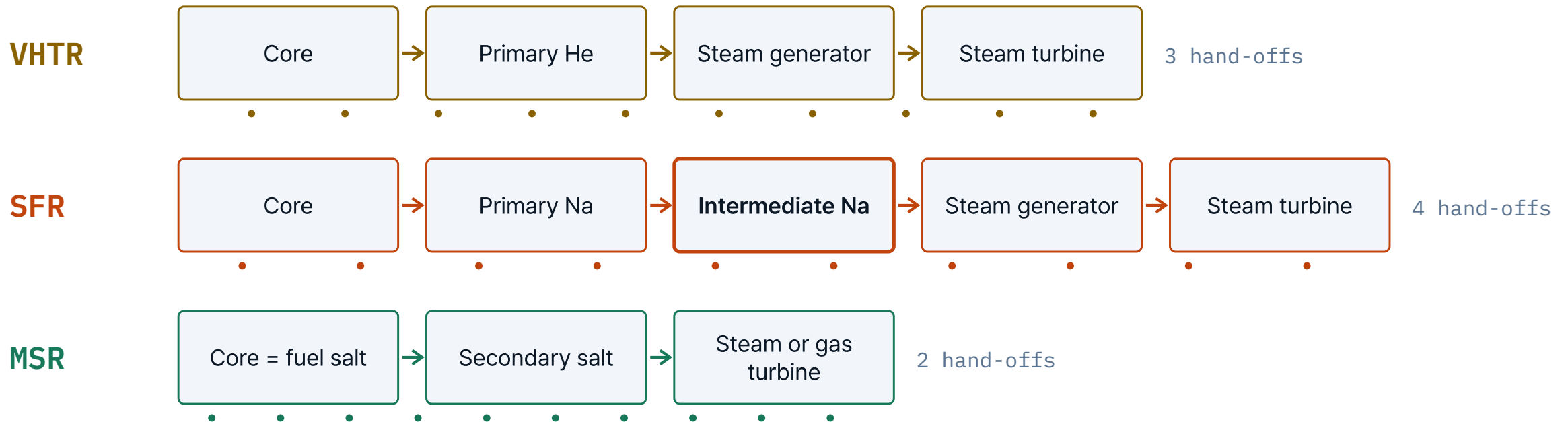
The three compared (2 of 2) - safety and maturity

- All three have negative temperature feedback **for different reasons** - Doppler, structural expansion, fuel expansion. **Void is positive only in a large SFR**

ITEM	VHTR (HTGR)	SFR	MSR
Void reactivity	essentially zero	positive in large cores	negative (salt expansion)
Temperature coeff.	negative (Doppler)	negative (structural expansion)	negative (fuel expansion)
Passive decay heat	conduction + radiation + RCCS	RVACS / DRACS	drain tank
Fuel limit	TRISO 1600 °C	FCCI 650-725 °C	none (liquid)
Refuelling	online (pebble) / shutdown (prism)	every few years, or none for 30 years	online
Hardest problem	air and water ingress, graphite life	sodium fires, sodium-water reaction, positive void	materials (not code-listed), source term, freezing
Commercial record	HTR-PM (2023-)	BN-600 (1980-), BN-800 (2016-)	none (TMSR-LF1, a 2 MWt test reactor)

Heat path by reactor - how many loops

- **Only the SFR adds an intermediate loop** - to keep activated primary sodium away from water. A sodium-water reaction then cannot reach the primary circuit
- In the MSR **the core is the fuel salt** - a clean secondary salt keeps it away from the power cycle



steam generator (SG) boils water with primary-side heat · **intermediate Na** a non-radioactive sodium loop between activated primary sodium and water · **secondary salt** a clean salt loop between fuel salt and the power cycle

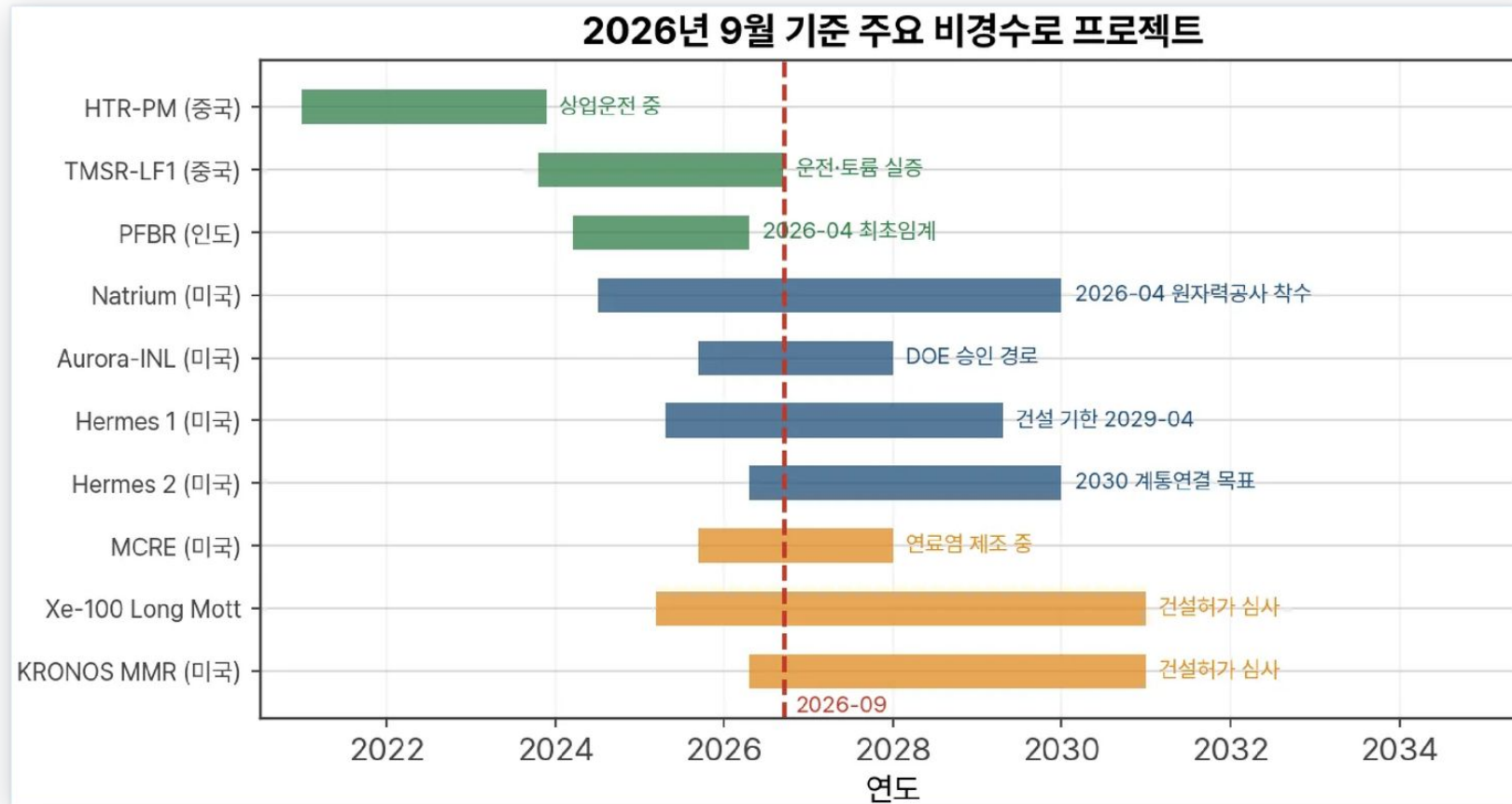
Source Redrawn from textbook Fig. 17.1 · the number of boxes is the number of fluids and components heat passes through · in the MSR the fuel salt is itself the primary fluid

Leader by axis - a different winner on each

- “Which reactor is furthest along?” has no single answer - **SFR** for operating record, **HTGR** for Gen-IV commercial and supply chain, **MSR** for demonstration and sea

AXIS	LEADER	BASIS
Commercial operation	SFR	BN-600 since 1980 - 46 years, 114 TWh
First Gen-IV commercial	HTGR	HTR-PM, December 2023, the world’s first
US licensing	SFR	Natrium - NRC construction permit March 2026, nuclear construction April
Fuel supply chain	HTGR	Baotou 300,000 pebbles a year; TRISO-X being built in the US
Demonstrator running	MSR	China’s TMSR-LF1, the only liquid-fuel reactor running
At sea	MSR	two Korean-led designs won class approval in 2026
Integral validation data	HTGR	the 2023 HTR-PM tests - the only commercial-scale loss-of-cooling data

Major non-LWR projects (September 2026)



Green = operating, blue = building or approved, orange = under review · the red line is September 2026 ·
textbook Fig. 17.2

AGES FASTEST

recheck every six months

① IAEA ARIS

**primary source for specs -
vendor documents**

② NRC APPLICATIONS

**US licensing, dated document by
document**

③ NEA SMR DASHBOARD

**the live online edition, not the
2023 PDF**

ARIS IAEA Advanced Reactors Information System - vendor-submitted specifications · NEA SMR Dashboard six-axis maturity assessment · PFBR India's 500 MWe prototype fast breeder · MCRE US Molten Chloride Reactor Experiment

Source Textbook Fig. 17.2 · as of September 2026 - the page that ages fastest · how to check: ch. 2 of *Finding SMR Data and PWR-type SMRs* (the weeks 3-4 deck)

Four slots to swap in a system code, per reactor

- Keep the core and junction structure; **swap only these four**. Effort runs **MSR > VHTR > SFR** - sodium needs properties and correlations, helium a multi-species model for air ingress, molten salt **more equations**

	Property class	Heat transfer · friction	Time step · numerics	Extra physics
VHTR (He)	ideal gas $P = \rho RT$	KTA 3102.2 / .3	sound 1880 m/s → implicit a must	solid conduction, radiation, multi-species diffusion
SFR (Na)	incompressible, linear $\rho(T)$	Mikityuk $Nu(Pe)$	low pressure, big inertia → long runs	axial conduction, pool stratification
MSR	salt tables (10-50 % uncertain)	reassess water correlations	freeze check $T < T_{\text{melt}}$	precursor transport, heat in the fluid

property class code object returning ρ , T and more from p and h · **implicit scheme** solves next-step values together - large time steps · **KTA 3102**

German pebble-bed heat-transfer and pressure-drop rules · **stratification** hot fluid floating on top, unmixed

Source Redrawn from textbook Fig. 17.3 · §17.5 · helium sound speed 1880 m/s at 750 °C (material Courant limit on 30 m/s flow)

System-code analysis compared by reactor

	VHTR	SFR	MSR
Property class	ideal gas (analytic derivatives)	incompressible (linear ρ -T)	salt tables (large uncertainty)
Two-phase model	not needed	not needed (normal operation)	not needed
Heat transfer	KTA 3102.2 (pebble) / Taylor (prism)	Mikityuk Nu-Pe	water correlations need reassessment
Friction	KTA 3102.3	standard	standard, but viscosity varies with T
Time-step limit	material Courant (30 m/s flow)	long transients	standard
Solid heat structures	dominant (graphite 73 $\times$ the fluid)	important (pool inertia)	secondary
Extra physics	multi-species diffusion (air ingress)	axial conduction, pool stratification	precursors, heat in fluid, freezing
Dedicated codes	GAMMA+ (KAERI), MELCOR	SAS4A/SASSYS-1, SAM	SAM, MARS-MSR
Integral validation	HTR-PM 2023, HTTR LOFC	EBR-II SHRT, Phénix	MSRE pump start-up

Courant limit fluid may not cross more than one cell per step · **GAMMA+** KAERI's HTGR code · **SAS4A/SASSYS-1** · **SAM** ANL's SFR and advanced-reactor codes · **SHRT** EBR-II shutdown heat removal tests · **LOFC** loss of forced cooling

Source This textbook, Table 17.4 · graphite heat capacity 73 $\times$ that of helium - ch. 5

Choosing a **reactor** by use

- All three are **at first-of-a-kind or earlier** - far behind LWR SMRs in how clear the licensing path is
- **Where THINK Lab fits** - all three lack TH tools and data, and the sea-going MSR adds hull motion. The emptiest cell is the best research topic

High-temperature process heat

HTGR

steel, hydrogen, petrochemicals. The only practical reactor for 750-950 °C - POSCO and Lotte Chemical are in the project

Burning spent fuel

SFR

no alternative when a fast spectrum is needed. Pairs with pyroprocessing

At sea, floating

MSR

atmospheric, single phase, gravity-driven shutdown, small core. The price is freezing and maintenance

What can be built now

LWR SMR

NuScale, i-SMR. The clearest licensing path

process heat heat itself supplied to industry, not electricity · **pyroprocessing** dry electrochemical separation of spent fuel in molten salt · **first-of-a-kind** the first unit of a design

Source This textbook §17.6 · for LWR SMRs (NuScale, i-SMR) see the weeks 3-4 deck

18

Wrap-up and next steps

What someone who has seen all three decks should be able to do - one sentence, five equations, a learning path, and the questions no one has answered yet.

The three decks in one sentence - **one cell** makes the reactor

- Change the coolant and one cell of the property table stands out - **and that cell drives every design decision and the hard problem**
- The order of analysis is the same for all three - **properties** → **heat transfer** → **layout** → **safety logic** → **tools**. Lead, heat pipes and sCO₂ yield to the same order

Helium
pcp 17



7 MPa

big circulator

passive safety on
graphite inertia

low power density

Sodium
Pr 0.005



new correlations

axial conduction

strong natural
circulation

reactivity →
intermediate loop

Molten salt
liquid fuel · 459 °C



precursor drift

heat in fluid,
online cleanup

off-gas source term

freezing

Five equations to keep

EQUATION	WHAT IT SAYS
$\dot{m} = Q / (c_p \Delta T)$	flow and temperature rise - helium's big circulator, salt's small pipes
$\Delta T_{\max} = q''' R^2 / (4k)$	the limit of conduction cooling - core diameter sets the power
$Nu = 0.047(1 - e^{-3.8(P/D-1)}) (Pe^{0.77} + 250)$	liquid-metal heat transfer - Pe instead of Pr
$\partial c_i / \partial t + \nabla \cdot (c_i \mathbf{v}_f) = (\beta_i / \Lambda) n - \lambda_i c_i$	precursors in circulating fuel - one advection term more
$P_d / P_0 = 0.0622 [t^{-0.2} - (t + 10^8)^{-0.2}]$	decay heat - where every passive-safety argument starts

q''' heat per unit volume · R core radius · P/D pitch ÷ pin diameter · Pe Péclet number = $Re \cdot Pr$ · $\mathbf{v}_f$ fuel-salt velocity · Λ neutron generation time

Source This textbook §18.2 · conduction - centre-to-surface difference of a cylindrical core (ch. 5) · Mikityuk correlation (ch. 7) · precursor transport (ch. 13) · Way-Wigner decay heat with 10^8 s of operation

A four-step learning path

- A 30-line calculation checks whether a slide's numbers are right - and you learn most where the solver refuses to converge

Step 1

Calculate it yourself

run the codes for §2.3
circulator power, §5.2
conduction cooling and §7.3
correlation comparison

Step 2

Put a reactor in the code

add the §6.3 helium or §11.3
sodium example to the Volume
III code and reach steady
state

Step 3

Solve a benchmark

① EBR-II SHRT-45R -
qualitative with our code ②
MSRE pump start - code
changes, clear physics ③
HTTR LOFC - solid heat
structures are key

Step 4

Check a primary source

pick one footnote and trace
it to the primary source -
practice for Finding SMR Data

Map of THINK Lab companion texts

WHEN YOU NEED	READ
SMR design requirements, defence in depth, ESFs	<i>SMR Design Requirements and ESFs - weeks 2-3</i>
Data sources (ARIS, NRC, NEA Dashboard), PWR SMRs	<i>Finding SMR Data and PWR-type SMRs - weeks 3-4</i>
Building a control-volume code	<i>Building a Reactor System Code I, II, III</i>
HTGR code, air ingress	<i>Introduction to GAMMA+</i>
Two-phase flow, flow regimes	<i>Two-Phase Flow and Flow Regimes, Flow Instabilities</i>
Designing scaled tests	<i>Scaling Methodology</i>
Risk assessment	<i>Introduction to PSA, HAZID Methodology</i>
Flow visualisation	<i>Index-Matched Flow Visualisation</i>

Ten common mistakes in presentations

- All are **caught by checking the primary source once** - re-quoting press articles and old slides is what spreads them

OFTEN SAID

ACTUALLY

HTR-PM runs at its rated 500 MWt	165 MWt per module, licensed to 200 - rising after the equilibrium core
Helium sound speed is ~1000 m/s	room temperature - 1882 m/s at 750 °C
Sodium $\rho = 1014 - 0.235T$ takes °C	kelvin - 7 % off in °C
4S is a loop design	pool (Toshiba's NRC submittal)
Oklo Aurora is a heat-pipe reactor	the 2020 design was dropped - now a sodium pool fast reactor
SEALER is sodium-cooled	lead-cooled
Seaborg's CMSR is NaOH-moderated	graphite since 2023 - and the company is now Saltfoss
ThorCon is 500 MWt	500 MWe (2 × 250 MWe)
USNC's MMR is 15 MWt	the old design - KRONOS as docketed is 45 MWth
The SFR limit is 883 °C sodium boiling	the real limit is the FCCI eutectic, 650-725 °C

FCCI fuel-cladding chemical interaction · MWt · MWe thermal and electric power · NaOH sodium hydroxide - once a moderator candidate

Source This textbook §18.5 · primary sources in the relevant chapters (2, 7, 9, 14, 5) · HTR-PM power as of May 2026

Questions this textbook could not answer

- For anyone looking for a research topic this may beat the main text - **MSR thermal hydraulics coupled to hull motion is the emptiest cell**

REACTOR	OPEN QUESTION
VHTR	effect of the -86 % irradiated-graphite conductivity uncertainty on accident results · wall bypass in porous-medium pebble-bed models · no Wigner-release data for graphite irradiated at HTGR temperature
SFR	no integral test where positive void really acts (blockage → local boiling) · 1-D treatment of upper-plenum stratification - where our visualisation fits · making up for an EM pump's missing inertia (coast-down design)
MSR	no system code with freeze/remelt · essentially no literature on MSR TH under hull motion · salt / cover-gas partition data (PIRT top) · redox-control failure in a PSA · drain-tank criticality when heeled
All	listing Hastelloy-N in ASME Sec. III Div. 5 - the code, not the technology, is the bottleneck · non-LWR source terms and EPZ sizing

dpa displacements per atom - a measure of irradiation damage · **Wigner energy** lattice energy stored in irradiated graphite · **EPZ** emergency planning zone · **ASME Div. 5** the high-temperature reactor component code

Source This textbook §18.6 · the -86 % graphite conductivity uncertainty rests mostly on data below 1 dpa · Hastelloy-N is not listed in ASME Sec. III Div. 5

Abbreviations (1 of 2) - reactors, fuel, reactor physics

Abbreviation	Expansion	What it means
MSR	Molten Salt Reactor	this deck: mostly liquid-fuel
MSRE	Molten Salt Reactor Experiment	ORNL's 8 MWt test reactor (1965-69)
MSBR	Molten Salt Breeder Reactor	ORNL's Th-U breeder concept
MSFR · MCFR	Molten Salt / Chloride Fast Reactor	European fast MSR · TerraPower chloride
FHR	Fluoride-salt High-temp. Reactor	salt-cooled, solid TRISO fuel (Kairos)
IMSR · CMSR	Integral / Compact MSR	Terrestrial Energy · Saltfoss (ex-Seaborg)
TMSR-LF1	Thorium MSR - Liquid Fuel 1	China's 2 MWt liquid-fuel test reactor
MCRE	Molten Chloride Reactor Experiment	US molten-chloride critical experiment
FLiBe · FLiNaK	LiF-BeF ₂ · LiF-NaF-KF	fluoride salts - FLiBe melts at 459 °C
LEU · HALEU	(High-Assay) Low-Enriched U	below 5 % · 5-20 % enrichment
Th-U · Pa-233	Thorium-uranium cycle	Th-232 → Pa-233 → U-233 conversion
β_{eff}	Effective delayed-neutron fraction	drops when fuel moves - 654 → 431 pcm
pcm · \$	per cent mille · dollar	10 ⁻⁵ Δk/k · reactivity over β _{eff}
U(IV)/U(III)	Redox ratio	redox index of the salt - governs corrosion
Hastelloy-N	—	Ni-Mo alloy - not listed in ASME Div. 5

Source This textbook, chs. 12-15 · the same entries as the abbreviation line under each slide

Abbreviations (2 of 2) - safety, sea, codes

Abbreviation	Expansion	What it means
PIRT	Phenomena Identification and Ranking	phenomena and importance ranking
PDC	Principal Design Criteria	NRC non-LWR criteria, RG 1.232
PSA	Probabilistic Safety Assessment	probabilistic safety assessment
EPZ	Emergency Planning Zone	emergency planning zone
ASME Div.5	ASME Sec. III Division 5	high-temperature reactor components
IMO · MSC	Int'l Maritime Org. · Safety Cttee	UN maritime body and its safety committee
SOLAS	Safety of Life at Sea	includes the nuclear-ship chapter
AiP · NTQ	Approval in Principle · New Tech. Qual.	class approval stages
ABS · LR · KR	—	US, UK and Korean class societies
KRISO	Korea Res. Inst. of Ships & Ocean Eng.	ship and ocean research institute
ORNL	Oak Ridge National Laboratory	ran MSRE, designed MSBR
SAM	System Analysis Module	ANL code - public MSRE model
Griffin	—	INL neutronics, coupled to SAM
MARS-MSR	—	KAERI code for MSRs - 25.5 s ripple
EDV	Emergency Depressurisation Valve	THINK Lab R134a tests

What is missing is not ideas but data

The design concepts for helium, sodium and molten salt were all on the table by the 1960s.
What is needed now is the experiments that test them, and the tools that can reproduce those experiments.

That is our work.